Differential CC $1\mu1\pi^{\pm}$ cross sections on argon in the NuMI beam

Inclusive measurement and the proton-tagged hadronic-mass / TKI subsample

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August 12, 2026

$\nu_\mu + \bar{\nu}_\mu$ charged-current single charged pion • FHC, RHC, combined

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Part I — Inclusive CC $1\mu 1\pi^{\pm} Xp$

Signal definition

$\nu_\mu/\bar{\nu}_\mu$ charged-current, exactly one muon and one charged pion, any number of protons, no π^0/kaons/heavier mesons; muon $p_\mu > 0.15$, pion $p_\pi > 0.175$ GeV/c, in the fiducial volume.

- Resonant $\Delta(1232)$ production dominates; probes pion production, nuclear effects and final-state interactions on argon.
- NuMI off-axis beam: ν_μ-dominant FHC, $\bar{\nu}_\mu$-enriched RHC.
- Five differential observables: p_μ, p_π, $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$.

Exposure (fake data, blind)

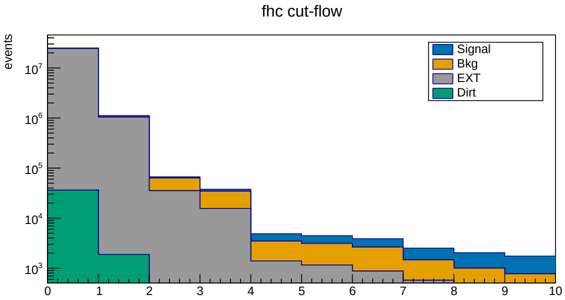
FHC (Runs 1,2,4,5)	8.86×10^{20} POT
RHC (Runs 1,2,3,4)	1.11×10^{21} POT
Combined	1.99×10^{21} POT

Event selection & cut-flow

- BDT-based $\mu/\pi/p$ particle ID.
- Vertex-in-FV, topology, track containment, muon/pion gap, π^0 shower veto, opening angle.
- Per-run POT scaling; EXT (beam-off) data-driven cosmics.

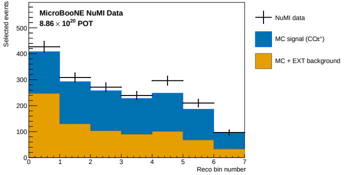
Final performance

Efficiency $\simeq$ 15.6%, purity $\simeq$ 55% (measured phase space).

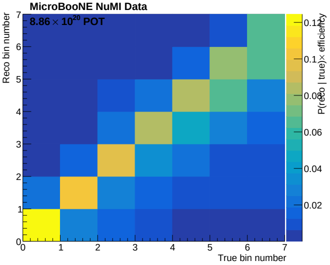


FHC cut-flow (signal blue, background orange, EXT grey, dirt green)

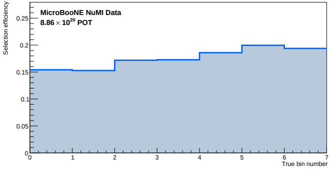
Reconstruction, response & efficiency



Reco p_μ (data/MC stack)



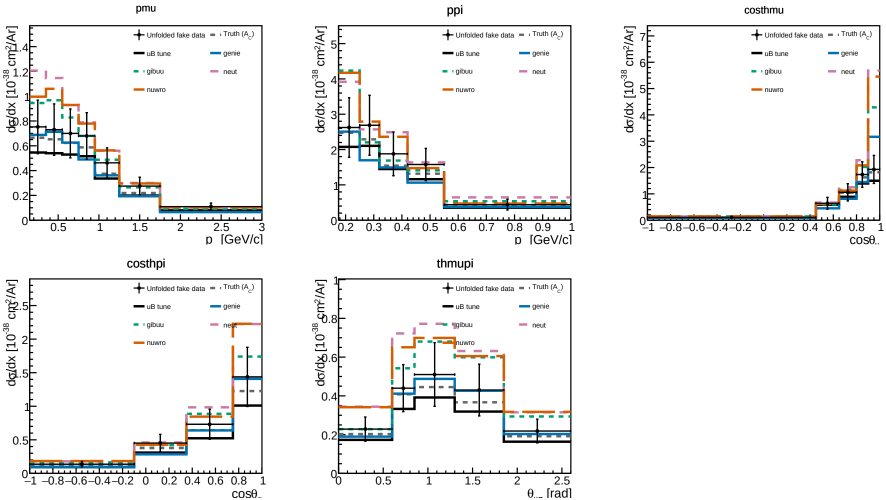
Response matrix



Efficiency vs p_μ

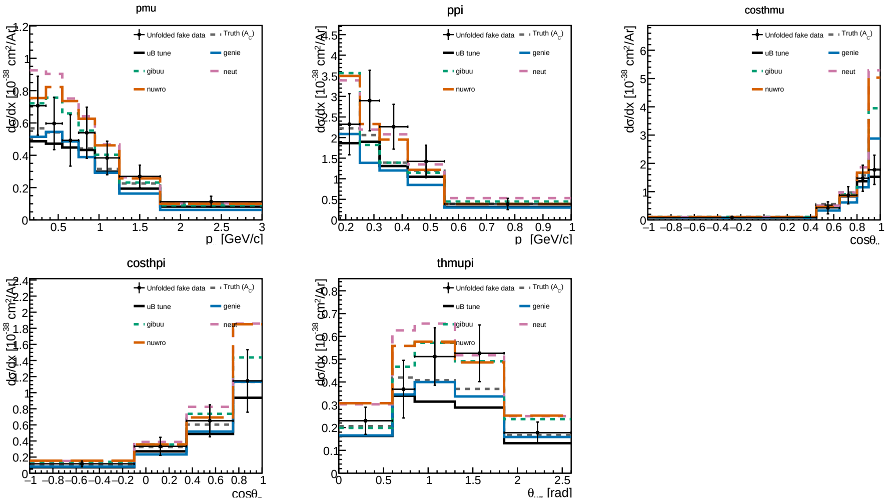
Wiener-SVD unfolding (framework `WienerSVDUnfolder`, 2nd-derivative regularisation) with the full systematic covariance from the universe throws.

Differential cross sections — FHC



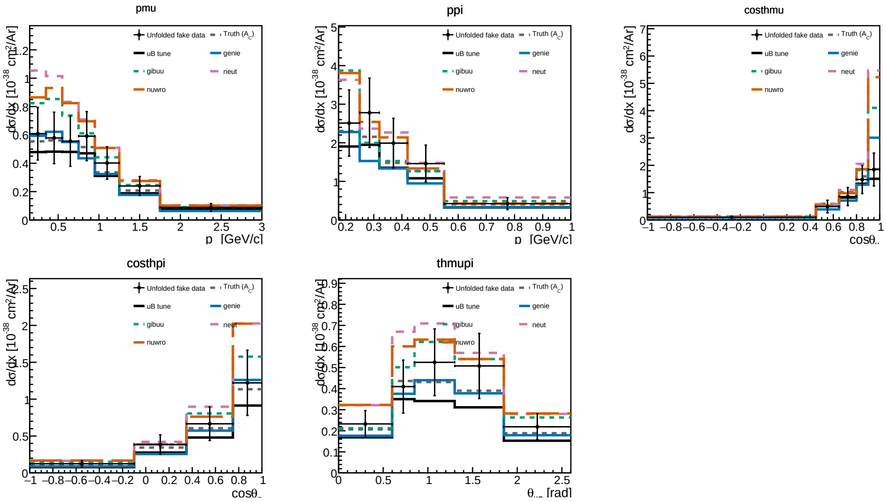
Unfolded fake data vs A_C-smeared truth, MicroBooNE tune, and GENIE/GiBUU/NEUT/NuWro.

Differential cross sections — RHC



$\bar{\nu}_\mu$-enriched beam; unfolded fake data vs A_C-smeared truth, MicroBooNE tune, and GENIE/GiBUU/NEUT/NuWro.

Differential cross sections — combined



FHC+RHC combined exposure (per-universe correlated flux treatment).

Integrated & total flux-averaged cross section

Integrated σ_{int} (Wiener-SVD)

	FHC	RHC	Comb
p_μ	0.99	0.86	0.82
p_π	0.98	0.96	0.96
$\cos\theta_\mu$	0.81	0.67	0.70
$\cos\theta_\pi$	0.98	0.80	0.86
$\theta_{\mu\pi}$	0.88	0.88	0.92

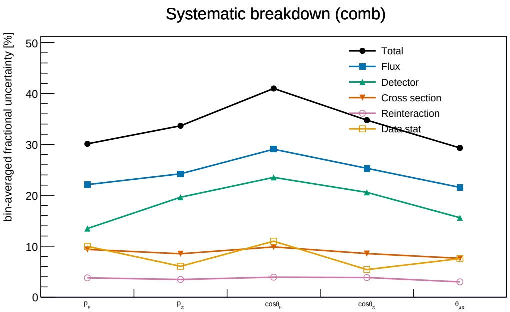
spread = Wiener-SVD A_C smearing (10^{-38}cm^2/Ar)

Total (cut-and-count, no unfolding)

Config	σ (10^{-38}cm^2/Ar)
FHC	1.06 ± 0.27
RHC	0.98 ± 0.35
Combined	1.02 ± 0.30

FHC generators: GENIE 0.79, GiBUU 1.08, NEUT 1.26, NuWro 1.20 — agree within $\sim 1\sigma$.

Systematics & sideband validation



Flux (PPFX) dominates; detector second; total $\sim$ 30–40%.

Background-control sidebands

Four signal-depleted regions (CC0π, multi-π, π^0, cosmic). High-statistics data/νMC $= 1.00$ across FHC/RHC/comb $\Rightarrow$ per-run weighting & POT machinery validated; ready to constrain the background at unblinding.

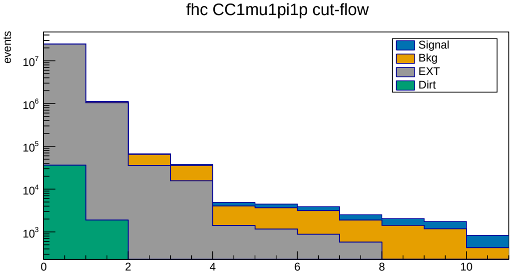
Part II — Proton-tagged subsample: hadronic mass & TKI

Idea

Requiring an identified proton reconstructs the full hadronic system, opening two new observable classes: the hadronic invariant mass (probes Δ/N^* directly) and the transverse kinematic imbalance (Fermi motion + FSI).

- `CC1mu1pi1p`: inherits the inclusive selection + a leading proton (LLR PID, $p_p \geq 0.3$ GeV/c, contained). Efficiency 11.5%, purity 48%.
- Six observables: $W_{\pi p}$, W_{had}, δp_T, $\delta\alpha_T$, $\delta\phi_T$, p_n (4 bins each).

Selection performance & proton-tag quality

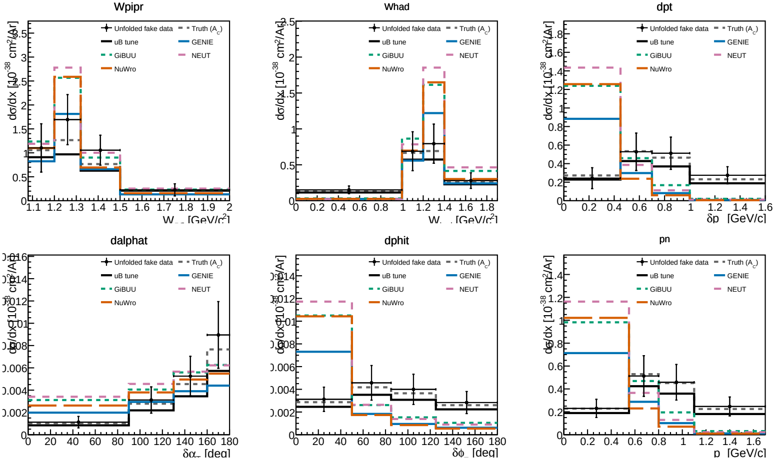


FHC proton-tagged cut-flow; last stage = proton identification

Proton tag (truth-matched, FHC)

- 87% tag purity ($\sim$ 13% mis-tag)
- Rejects $\sim$ 64% background for $\sim$ 30% signal loss
- Purity 33% $\rightarrow$ 48%; $S/\sqrt{B}$ +17%
- Proton p: -9% bias, 21% resolution

Proton-tagged differential cross sections (FHC)



Unfolded fake data vs truth, uB tune, and all four generators. Closure $\chi^2/\mathrm{ndf} \ll 1$ for every observable.

Generator comparison & background composition

Integrated proton-tagged σ $(10^{-38}\mathrm{cm}^2/\mathrm{Ar})$

GENIE 0.50 $<$ NuWro 0.65 $<$ GiBUU 0.74 $<$ NEUT 0.78; data $\sim$ 0.58.
Spread $\sim 1.6\times$; largest in low-δp_T/high-p_n (FSI-sensitive).

NuWro/NEUT reprocessed from event vectors in the SL7 container.

Remaining background (FHC)

Component	%
ν_μ CC wrong topology	64
True CC$1\mu1\pi$, no p>0.3	18
NC	11
Out-of-FV	5
ν_e CC	1

82% true ν_μ CC — well modelled & constrainable.

Summary

- Inclusive: differential CC$1\mu1\pi^{\pm}$ cross sections for FHC, RHC and combined, per-run POT scaling, Wiener-SVD unfolding; total flux-averaged σ agrees with GENIE/GiBUU/NEUT/NuWro within $\sim 1\sigma$.
- Proton-tagged: new hadronic-mass ($W_{\pi p}$, W_{had}) and TKI (δp_T, $\delta\alpha_T$, $\delta\phi_T$, p_n) observables; FHC closure validated; all four generators compared and bracket the data.
- Blind analysis: fake data throughout; sidebands ready to constrain the background before unblinding.
- Ongoing: RHC/combined W/TKI in production; detector systematic and a tighter-proton-PID study underway.

Thank you — questions?

Backup

Backup: per-run exposure

Mode	Run	Data POT	MC/data scale
FHC	1	3.28×10^{20}	0.141
	2	1.27×10^{20}	0.051
	4	2.08×10^{20}	0.073
	5	2.23×10^{20}	0.116
RHC	1–4	1.11×10^{21}	0.045–0.091

Each run scaled by its own $D_{\mathrm{run}}/\mathrm{MC}_{\mathrm{run}}$ (real-data-ready).

Backup: systematic breakdown (combined)

Source	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$
Total	30.1	33.7	41.0	34.8	29.3
Flux (PPFX)	22.1	24.2	29.1	25.3	21.6
Detector	13.5	19.6	23.5	20.6	15.6
Cross section	9.4	8.5	9.9	8.6	7.6
Data stat	10.0	6.1	11.0	5.4	7.6

Bin-averaged fractional uncertainty [%]. Flux dominates throughout.

Backup: W/TKI observable definitions

From the muon, charged-pion and leading-proton momenta (STV tool, option 1):

$$W_{\pi p} = \sqrt{(E_\pi + E_p)^2 - |\vec{p}_\pi + \vec{p}_p|^2} \quad \text{(direct, no } \nu \text{ direction)}$$

$$W_{\text{had}} = \sqrt{m_p^2 + 2m_p(E_{\text{cal}} - E_\mu) - Q^2} \quad \text{(calorimetric)}$$

$$\delta p_T = |\vec{p}_T^{\,\mu} + \vec{p}_T^{\,\text{had}}|, \quad \delta\alpha_T,\ \delta\phi_T \text{ (TKI angles)}, \quad p_n = \sqrt{\delta p_T^2 + p_L^2}$$

Proton momentum from the proton-hypothesis range kinetic energy; $E_b = 24.78$ MeV argon removal energy.

Backup: proton-tagged binning & closure

Observable	Bin edges
$W_{\pi p}$	1.08, 1.20, 1.32, 1.50, 2.00 GeV/c^2
W_{had}	0, 1.00, 1.20, 1.40, 1.90 GeV/c^2
δp_T	0, 0.45, 0.70, 1.00, 1.60 GeV/c
$\delta \alpha_T$	0, 90, 130, 160, 180 deg
$\delta \phi_T$	0, 50, 85, 125, 180 deg
p_n	0, 0.55, 0.80, 1.10, 1.70 GeV/c

FHC closure σ_{int}: 0.64, 0.59, 0.56, 0.56, 0.63, 0.54; $\chi^2/\mathrm{ndf} = 1.5, 0.2, 0.4, 0.5, 0.3, 0.1$ over 4 bins.

Backup: generator predictions

- Inclusive: per-observable $d\sigma/dx$ from GENIE (gst), GiBUU (`FinalEvents`), NEUT (neutvect), NuWro (event output), flux-combined to per-argon 10^{-38} cm^2.
- Proton-tagged W/TKI: same event samples reprocessed with the proton tag + TKI construction; NuWro and NEUT run inside the SL7 apptainer container.
- FTE rebinned to the coarsened analysis binning so every generator overlays on every observable.